Issue	**1**
Factory ID	**26310**
Date of Issue	**13-Apr-2026**

A A Garments

Chak 220 RB, Street 01, Block A, Babu Wala, Jhang Road
38001 Faisalabad
Punjab Pakistan

Latitude: [31.4137], longitude: [72.9984]
Date of inspection: 09-Mar-2026

Electrical Safety Inspection Report

Observations & Actions



Authors	**Zeeshan Haider**
Reviewed by	**Majid Khan**
Approved by	**Paul Rigby**

Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 09-03-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 09-Mar-2026, Syed Hassan Hussain and Zeeshan Haider on behalf of the Pakistan Accord carried out an initial electrical inspection for A A Garments, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Wazeer (Compliance Adviser), and Mr. Shahid Alve (Manager Export), along with other staff members of the factory. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection, other representatives from Pakistan Accord were also present.

The factory comprises four buildings, details of which are provided below:

1. **Main Building** - 3 storeys (Basement + Ground + 1). The building is used for a fabric store and cutting (basement), checking and packing (ground floor), stitching (first floor), solar panels (rooftop). The building was constructed in 2022.

2. **Office Building** - is a 2-storeys (Ground+1). The building is used for offices (ground and first floor). The building was constructed in 2022.

3. **Utility Building** is a 2 storey (Ground + 1). The building is used for canteen (ground floor), 1st floor is not occupied. The construction date of building is 2022.

4. **Canteen Building** is a 2 storey (Ground + 1). The building is used for utilities (ground floor) and prayer area (1st floor). The construction date of building is 2022.

A summary of the main concerns in relation of life safety systems identified during the inspection (non-exhaustive) that should be actioned are as follows:

Hot Spots: Hotspots were found exceeding safe limits in many distribution boards within the facility. Significantly increases the risk of insulation failure, equipment damage, arc flashes, and fire outbreaks. They may also lead to unexpected power outages, operational disruptions, and potential injury to personnel. Immediate actions are required to resolve this issue.

Earthing System: The earthing system was non-compliant with the standard due to inaccessible earth pits and undersized main earth cables. In addition, bonding of cable trays, metal structures, doors, and metallic furniture/fittings were absent. These pose significant safety risks, such as electric shock, fire hazards, and potential equipment damage.

We see no reason to suspend operations in the facility due to these concerns, subject to the action identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Site plan (Satellite view) - Red dashed line indicates site boundary highlighted (blue) area indicates building inspected.

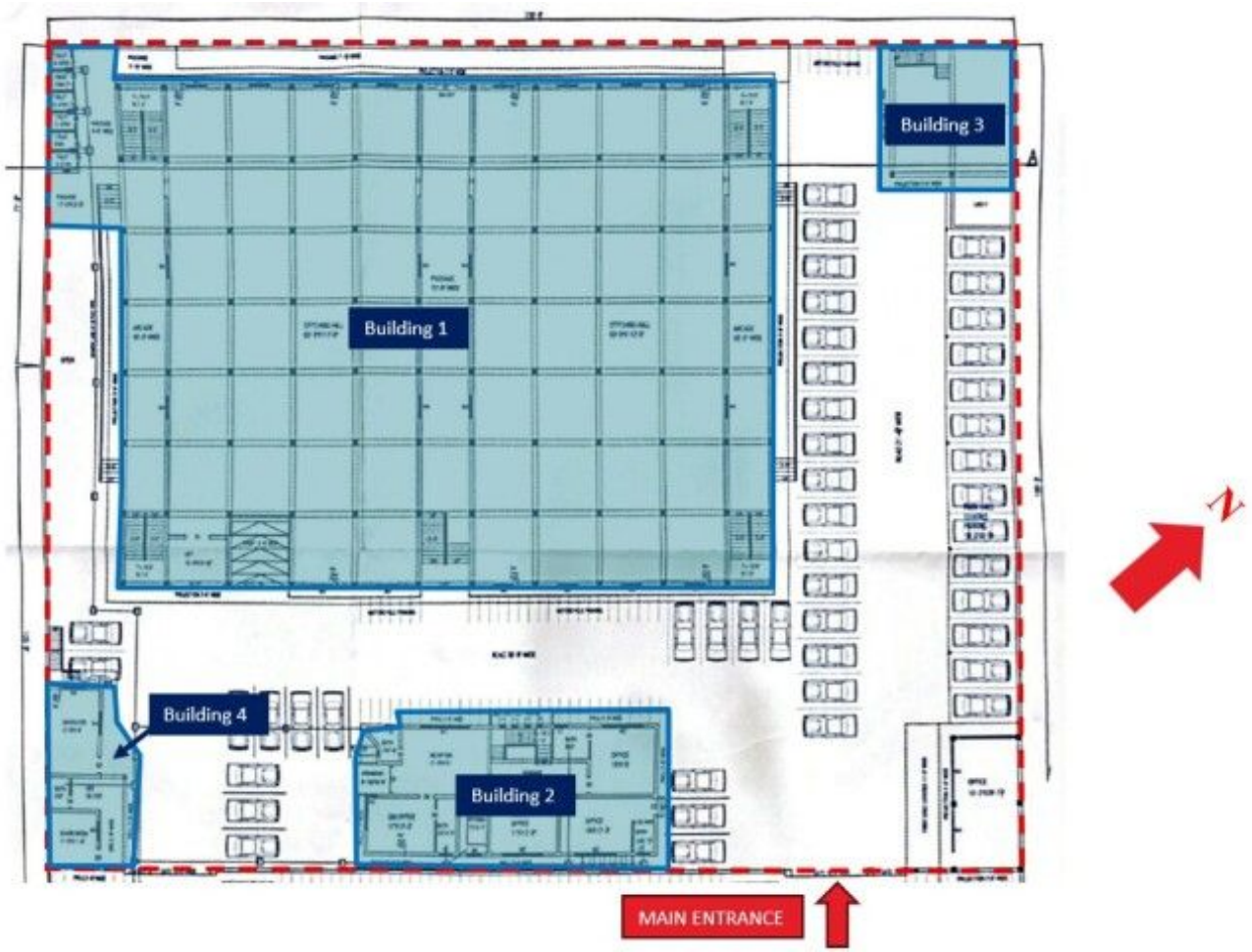


Figure 1.2. Site layout. Red dashed line indicates site boundary, highlighted area (blue) indicates buildings inspected.

2.1 Building Use

The factory comprises the following buildings:

Main Building:
B00 - Basement: Fabric store and cutting.
L00 - Ground floor: Checking and packing.
L01 - First floor: Stitching.
L99 - Rooftop: Solar panels.

Office Building:
L00 - Ground floor: Canteen.
L01 - First floor: Not occupied.

Utility Building:
L00 - Ground floor: Utilities.
L01 - First floor: Prayer area.

Canteen Building:
L00 - Ground floor: Offices.
L01 - First floor: Offices.

2.2 Utility Source

The factory receives low-voltage electrical supply from WAPDA (sanctioned load 69 kW), which is fed into the main LT panels, and then low-voltage power is distributed to the respective buildings floors distribution boards. An additional 80 kW on-grid solar system was also installed at rooftop.



Figure 1.3. 250 kVA PMT for factory primary power source

2.3 Backup source

The factory has one diesel generator rated at 100 kVA, which serves as a backup power source. Additionally, the emergency lighting and fire alarm systems are equipped with built-in battery backup.



Figure 1.4. 100 kVA diesel generator for backup power source

3. Documentation

This table shows the details of the documentation provided by the factory:

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	No	-	-
2	Single Line Diagrams	No	-	-
3	Fire Alarm Drawing	No	-	-
4	Earth Test Report	No	-	-
5	Thermographic Test Report	No	-	-
6	Insulation Resistance Test Report	No	-	-
7	LPS Risk Assessments	No	-	-
8	Periodic Maintenance Record	No	-	-
9	Transformer Oil Testing Report	No	-	-

4. Electrical Safety Measures

During our electrical site inspection of A A Garments, a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within A A Garments. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a trained professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202603-063862

Timeline	Remediation Plan
Medium Term CAP-0074651	Factories should begin remediation of CAP items such as cleaning, cable organisation, and basic corrections using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no electrical room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.
The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and an increased likelihood of equipment failure. F-202603-063851

Timeline	Remediation Plan
Medium Term CAP-0074656	Develop detailed electrical room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0074657	Regularly update and review the electrical room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were not available.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack the critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202603-063853

Timeline	Remediation Plan
Medium Term CAP-0074658	Develope comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0074659	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was observed that the factory does not have available electrical layout drawings, including power layouts, lighting layouts, fire alarm layouts, earthing layouts and containment layouts.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202603-063852

Timeline	Remediation Plan
Medium Term CAP-0074660	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0074661	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.
During the survey, the circuit breakers installed throughout the facility inside the electrical panels and distribution boards were inspected.

i. It was found that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes such as:
a) Main LT panels 160A circuit breaker with 25sqmm cable.
b) Main Building basement distribution board 225A circuit breaker with 25sqmm cable .

Circuit breakers that are not properly sized may fail to trip during overload conditions, allowing excessive current to flow and causing overheating, which leads to a fire risk.

ii. It was also found at multiple locations that single-phase outgoing circuit was taken from three pole Breaker. Using a single phase of a three-pole breaker can lead to unbalanced load conditions. This can cause the breaker to trip unexpectedly or, in some cases, fail to trip when necessary, compromising the safety of the system. This was observed at ground floor main LT panels. F-202603-063840

Timeline	Remediation Plan
Medium Term CAP-0074675	i. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.
Medium Term CAP-0074676	ii. Remove the single-phase load from the three-pole breaker and use a single-pole breaker for single-phase connection out as per standard cable current carrying capacity.



Figure 2.1 - Inadequate generator breaker rating at main LT panels 300A circuit breaker with 25sqmm cable



Figure 2.2 - Single phase taken out of TP MCCB in the LT panel



Figure 2.3 - Inadequate breaker rating at main building basement distribution board 225A circuit breaker with 25sqmm cable

7.2 Hot Spots

This relates to the entire facility.
During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey detected several hot spots in multiple distribution boards, observed at:
a). Main building ground floor distribution boards (Temperature 161 °C).
b). Main building first floor distribution boards (Temperature 69 °C).
c). Main building basement area distribution boards (Temperature 58.1 °C).

Hot spots in electrical distribution systems can be caused by several factors, including:
> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation
All temperature related assessments are based on Table 100.18 of ANSI Standard 2019.

It was also found that the thermographic test report was not available at the site.

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment.F-202603-063846

Timeline	Remediation Plan

Immediate CAP-0074694	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term CAP-0074695	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0074696	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating. Submit the thermographic test report to the Pakistan Accord for review.



Figure 2.4 - Hot spot was found at main building ground floor distribution board



Figure 2.5 - Hot spot was found at main building ground floor distribution board



Figure 2.6 - Hot spot was found at main building first floor distribution board

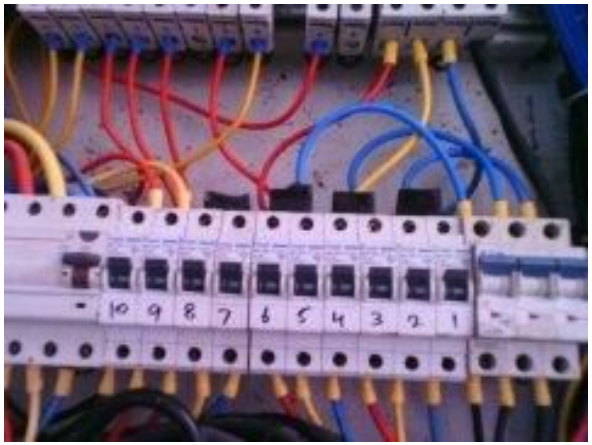


Figure 2.7 - Hot spot was found at main building first floor distribution board



Figure 2.8 - Hot spot was found at main building basement area distribution board



Figure 2.9 - Hot spot was found at main building basement area distribution board

7.3 Earthing system

This section relates to the entire facility.
During the inspection of the facility's electrical panels, distribution boards, and machinery, the following observations with the earthing system were identified.

i. It was observed that the factory does not have an earthing layout drawing available.

ii. It was found that the earth testing record was not available at the site. Without earth testing records, it is difficult to ensure that the grounding system is functioning correctly and that maintenance activities.

iii. It was also found that the equipotential bonding for the distribution board front doors, cable trays, machines body, metallic cage of generator and panel covers were missing.
Observed at,
a) Main building ground and first floor distribution boards.
b) Office building ground and first floor distribution boards.

iv. The earth pits were inaccessible and unsuitably located in LT room and outdoor area of the facility.

v. It was noted that the earth cables were not appropriately sized according to the relevant earthing standards.
Observed at
a) Ground floor main LT panels and floor distribution boards of the main building.

vi. It was observed that the power plugs used for the factory’s main building stitching machines were mismatched with the installed switch boards, due to which the earth pin was not properly connected. As a result, earth continuity was not maintained for the machine.

The lack of proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202603-063839

Timeline	Remediation Plan
Medium Term CAP-0074667	i. Develop and submit the earthing system design to the Pakistan Accord for review.

Medium Term CAP-0074668	ii. Prepare and submit the earth testing record to the Pakistan Accord for review.
Longer Term CAP-0074669	iii. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Longer Term CAP-0074670	iv. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.
Longer Term CAP-0074671	v. Replace all earthing cables with appropriately sized cables as per the requirements of BS 7671 18th Edition + A2 (2022) Appendix 4, Section 5. Verify that all earthing conductors meet the minimum cross-sectional area specified.
Longer Term CAP-0074672	vi. The mismatched plugs and switch boards should be replaced with properly compatible and standardized industrial plugs and sockets that ensure proper earth pin connection.
Longer Term CAP-0074673	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.10 - Earth distribution and bonding connection were missing inside main building first floor lighting distribution board



Figure 2.12 - Power plug used with multipin socket which does not connect earth conductor for all stitching machines of first floor main building



Figure 2.11 - Earth pits were found inaccessible at ground floor LT room



Figure 2.13 - Multipin power board used for main building first floor stitching machines power



Figure 2.14 - Earth distribution and bonding connection were missing inside main building ground floor lighting distribution board

7.4 Panel/cabling/busbar

This section relates to the entire facility.
The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.

i. It was also found that there was loose wiring and insufficient working space inside the distribution boards for operation and maintenance. Loose wiring and insufficient space, lead to excessive heat generation which can cause the insulation to melt and potentially lead to fire.
Observed at office building ground floor distribution board.

ii. PVC flexible pipe was installed inside the few distribution boards, causing overheating risk, poor ventilation, and non-compliance with electrical standards.
Observed at,
a) Main building ground floor distribution board.
b) Office building first floor distribution board.F-202603-063841

Timeline	Remediation Plan
Medium Term CAP-0074681	i. Install wire ducts to organize and route wires neatly, remove loose cables, and ensure that no cables are attached to the bus bars. This may mean the addition of new distribution boards with proper accessibility for cables incoming and outgoing, bus-bar alignment.
Medium Term CAP-0074682	ii. Remove the PVC pipe and use proper cable management (e.g., ducts or trunking) suitable for panel use to ensure safety, heat dissipation, and compliance.
Longer Term CAP-0074683	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.15 - PVC flexible pipe was found inside main building ground floor distribution board



Figure 2.17 - Inadequate cable dressing was found at ground floor office building distribution board



Figure 2.16 - PVC Flexible pipe was found at office building first floor distribution board

7.5 Loop Connection

This section relates to the entire facility.

During the inspection loop connections were identified within the electrical distribution boards.
In these loop connections, one circuit breaker is energized by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.
Observed at,
a) Main building basement, ground and first floor distribution boards.
b) Office building ground and first floor distribution boards.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202603-063844

Timeline	Remediation Plan
Medium Term CAP-0074690	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilised to provide each circuit with the required live, neutral, and earth connections.



Figure 2.18 - Loop connections were found at main building basement distribution board



Figure 2.19 - Loop connections were found at main building ground floor distribution board

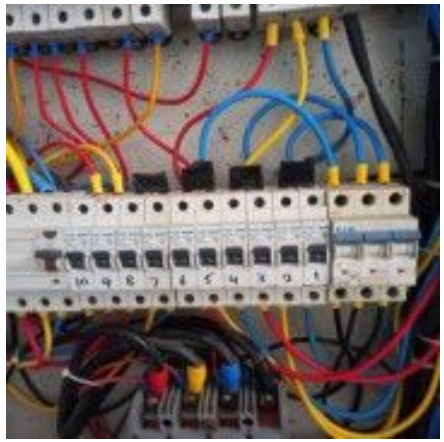


Figure 2.20 - Loop connections were found at main building first floor distribution board

7.6 Multiple Connections

This section relates to the entire facility.

During our inspection, multi looping was observed in a number of electrical panels across the facility. In these cases, multiple circuits were being protected by a single circuit breaker.
Observed at,
a) Main building ground and first floor distribution boards.
b) Office building first floor distribution board.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit. F-202603-063842

Timeline	Remediation Plan
Medium Term CAP-0074687	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022).

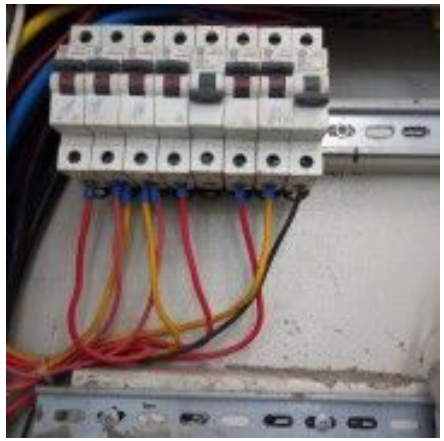


Figure 2.21 - Multiple connections were found at main building basement distribution board

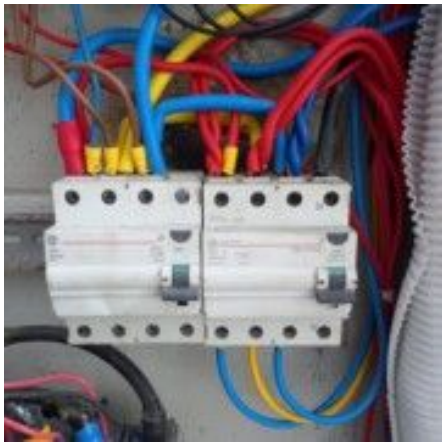


Figure 2.22 - Multiple connections were found at main building ground floor distribution board



Figure 2.23 - Multiple connections were found at office building first floor distribution board

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

It was found that many distribution boards have openings, which may cause potential hazards due to environmental factors.
Observed at,
a) Main Building ground floor LT panels.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202603-063843

Timeline	Remediation Plan
Medium Term CAP-0074688	Examine the entire facility panels to identify all openings, holes, and gaps that need to be sealed. Seal the openings, holes of panels, cable trays and trenches with metallic enclosures, use cable glands for cable entry. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0074689	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.24 - Unsealed opening was found inside main building ground floor LT panels



Figure 2.25 - Unsealed opening was found inside main building ground floor LT panels

7.8 Adequate accessibility

This section relates to the entire facility.
During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At multiple locations, it was found that the electrical distribution boards do not possess the required space for inspection or easy access to the panels and their components.
Observed at,
a) Main building rooftop RO plant room distribution board.

Limited accessibility to distribution boards and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202603-063845

Timeline	Remediation Plan
Medium Term CAP-0074693	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.26 - Inadequate accessibility was found for main building rooftop RO plant room distribution board

7.9 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

It was found that the lightning protection system was provided, however, there was no study available to confirm its adequacy.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202603-063849

Timeline	Remediation Plan
Medium Term CAP-0074691	Hire a qualified engineer to perform risk assessment according to BS 62305. This study should evaluate whether the current system meets the necessary design and performance standards.
Longer Term CAP-0074692	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant standards such as BS EN 62305 (Protection against Lightning).



Figure 2.27 - Lightning arrestor was installed at the main building rooftop

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section relates to the entire facility.
During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

It was observed that conventional and not listed fire alarm systems was installed for the whole facility. However, several deficiencies were noted including the absence of fire alarm devices in offices, utility rooms, corridors, the use of non-listed devices, insufficient audibility and the lack of visual alarms. Detectors were installed not at the highest point. Additionally, fire-rated cables are installed throughout the system. However, no documentation is available to verify their listing or certification.

Non-listed fire alarm panels may not meet standards and regulatory requirements, leading to potential issues with reliability and performance. Conventional fire alarm systems may lack advanced features such as individual device identification and precise location of alarms, which can delay response times during emergencies.
Non-functional, Incomplete fire alarm coverage and lack of listed fire-rated cable and detectors installed on the beam may increase the risk of delayed or missed detection of fire incidents, potentially leading to major damage.F-202603-063848

Timeline	Remediation Plan
Medium Term CAP-0074700	Ensure the functionality and performance of the existing alarm system within facility.
Medium Term CAP-0074701	(i) Submit design documentation for new replacement system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets and product certifications of all listed equipment to show full compliance of the Fire Alarm and Detection System (FADS) with NFPA 72.
Medium Term CAP-0074702	(ii) Submit documentation to Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.
Medium Term CAP-0074703	(iii) Submit documentation to Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on site for installation.

Longer Term CAP-0074704	(iv) Once installation of the FADS system is complete, factory to submit full "as-built" documentation of the system to Pakistan Accord for review.
Longer Term CAP-0074705	(v) Factory to advise Pakistan Accord when FADS installation fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0074706	(vi) Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.28 - Conventional and non listed fire alarm panel installed for the facility



Figure 2.29 - Fire detector was found installed above the ceiling fan at ground floor production area



Figure 2.30 - No fire detection device was found in LT room



Figure 2.32 - No fire detection device was found at main building first floor corridor areas



Figure 2.31 - Non fire rated cable used for MCP at main gate LT room



Figure 2.33 -



Figure 2.34 - No fire detection device was found at main building first floor corridor areas

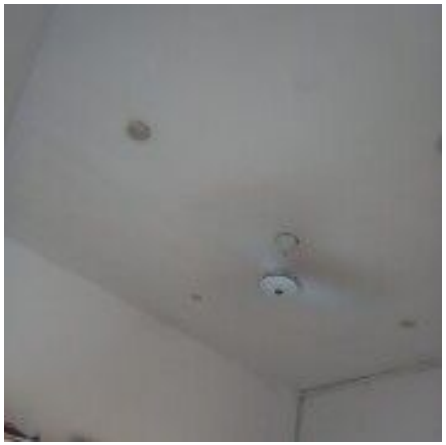


Figure 2.35 - No fire detection device was found at main gate fire control room

8.2 Maintenance records

This section refers to the entire facility.

During the inspection, the testing and maintenance records for the FADS were reviewed.

It was found that the routine testing of the Fire Alarm and Detection System (FADS) has not been performed consistently.

Routine testing and maintenance of the Fire Alarm and Detection System (FADS) in a factory are essential to ensure the system's reliability and functionality, enabling early fire detection and preventing unexpected failures. It also ensures compliance with NFPA 72 and other regulatory standards, enhances occupant safety through effective evacuation, and minimises operational downtime and costs.F-202603-063854

Timeline	Remediation Plan
Longer Term CAP-0074699	Conduct regular testing of the Fire Alarm and Detection System (FADS) to verify proper operation under both mains and backup power supply conditions. Ensure any faults indicated on the FADS control panel are promptly diagnosed and rectified. Maintain accurate records of all routine test results to ensure ongoing compliance and system reliability in accordance with Chapter 14 of NFPA 72.

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection it was observed that emergency lighting was provided but not functional in a number of locations such as:
a) Main building ground and first floor production area.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202603-063856

Timeline	Remediation Plan
Medium Term CAP-0074715	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.
Longer Term CAP-0074716	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.36 - Emergency light was found non-functional at main building ground floor production area

9.2 Provided but insufficient

The section refers to the entire facility.
During the inspection the emergency lighting arrangement throughout the facility was assessed.
The following observations were made:

i. It was observed that the emergency lighting layout drawing was not available at the site.

ii. It was also observed that the emergency lighting maintenance record was not available.

iii. During the inspection, it was found that the emergency lights and signage provided in the facility were insufficient to cover the entire facility, leading to dark or poorly lit areas during power outages. This inadequate coverage may cause incidents during emergencies.
Observed at,
a) Main building all floors production area, offices, warehouses, stores, electric rooms, corridor area, and stairs.
b) Office building ground and first floor offices, stairs.

iv. External emergency lighting was not provided for the facility, preventing people from reaching a place of safety outside the facility during nighttime evacuations. Observed at main and office building outdoor area.

v. The backup power supply to the emergency luminaires is insufficient in different areas such as main building ground floor area emergency lights.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202603-063847

Timeline	Remediation Plan
Medium Term CAP-0074708	i. Submit the emergency lighting system design to the Pakistan Accord for review.
Longer Term CAP-0075859	ii. Submit the emergency lighting system maintenance record to the Pakistan Accord for review.

Medium Term CAP-0074709	iii. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Install additional emergency luminaires in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation.
Medium Term CAP-0074710	iv. Install external emergency luminaires to enable safe evacuation during hours of darkness all the way to a safe place outside the building(s).
Longer Term CAP-0074711	v. Upgrade the backup power supply to ensure it can adequately support all emergency luminaires during a power failure. This includes installing additional battery packs or generators as necessary to comply with the Building Code of Pakistan Fire Safety Provisions 2016.
Longer Term CAP-0074712	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.37 - Insufficient emergency lights and signage



Figure 2.38 - Insufficient emergency lights and signage were found at

were found at main building basement production area



Figure 2.39 - Emergency light was found missing in main building production area stairs



Figure 2.41 - Emergency light was found missing in main building first floor stairs

main building first floor production area

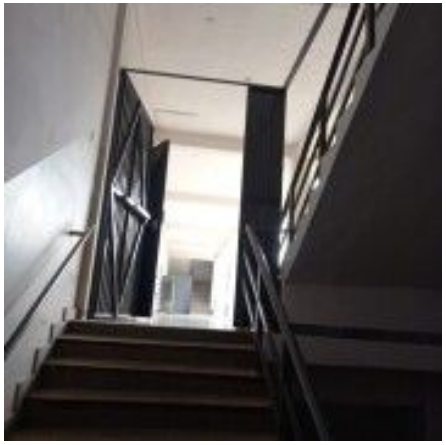


Figure 2.40 - Emergency light was found missing in main building ground floor stairs



Figure 2.42 - Emergency light was found missing in main building first floor corridor area



Figure 2.43 - Emergency light was found missing in office building office rooms



Figure 2.44 - Emergency light was found missing in office building stairs

9.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further the emergency lights and exit signs be reconfigured to be connected on an independent single circuit, which must not include any other load except for the emergency lighting system according to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202603-063850

Timeline	Remediation Plan
Medium Term CAP-0074713	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately when general lighting power fails, as required by the Building Code of Pakistan Fire Safety Provisions 2016.
Longer Term CAP-0074714	Provide an independent emergency light circuit as required by the Building Code of Pakistan Fire Safety Provisions 2016. Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

10. Generator, Transformer & Sub-Station Room

10.1 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

i. During the inspection, it was found that the battery terminals were exposed, Corroded and not properly insulated, bare terminals increase the risk of electrical shock or short circuits. potentially igniting fires or explosions. Proper monthly and annual maintenance of the battery system is crucial for fire safety.
Observed at the ground floor generator and electric room.

ii. The placement of some batteries did not comply with safety regulations such as in office building near stairs, potentially hindering safe access and proper ventilation.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202603-063855

Timeline	Remediation Plan
Medium Term CAP-0074717	i. Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term CAP-0074718	i. Establish a routine inspection and maintenance protocol to regularly check and maintain proper insulation of all battery terminals.
Medium Term CAP-0074719	ii. Reposition batteries that are improperly placed to ensure they comply with safety regulations, allowing for safe access and adequate ventilation. Mark and clear pathways around battery installations to ensure easy and safe access for maintenance and emergencies.
Longer Term CAP-0074720	ii. Conduct regular audits of battery placements to ensure ongoing compliance with safety standards and regulations.



Figure 2.45 - Generator battery terminals were found corroded and lacking insulation



Figure 2.46 - Electric room battery terminals were found lacking insulation



Figure 2.47 - Inappropriate placement of batteries were found at office building first floor near stairs

11. Cabling & Wiring

11.1 Cable support and protection

This section relates to the entire facility.
During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

It was observed that cables were installed without proper containment systems such as cable trays, conduits, or ducting, and were laid directly on the floor. The absence of adequate cable containment and support can cause sagging or mechanical stress on the cables, potentially leading to damage or failure.
Observed at,
a) Ground floor LT room generator cable.
b) Main building rooftop lift room cable.

Cables on the floor are susceptible to mechanical damage from foot traffic, material handling equipment, and other operations, which could result in electrical short circuits, potential electric shock, or even fire. Additionally, the exposed cables can cause workers to trip and fall, leading to possible injuries.F-202603-063836

Timeline	Remediation Plan
Medium Term CAP-0074725	Secure cables with appropriate containment and supports to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations).
Longer Term CAP-0074726	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.



Figure 2.48 - Generator cable was found without containment at main building ground floor LT room



Figure 2.49 - Cable was found without containment at main building rooftop lift room

11.2 Splicing of cables

This section relates to the entire facility.
During the inspection, the cable installation was visually surveyed throughout the facility. The following observations were noted:

i. It was found that cable terminations were made without using cable lugs, leading to broken strands and loose connections.
Observed at,
a) Main building basement, ground and first floor distribution boards.

ii. PVC tape joints were found at multiple locations. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical short circuits and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks.
Observed at,
a) Main building basement, ground and first floor distribution boards.
b) Office building ground and first floor distribution boards.

iii. It was noted that the insulation of main building basement cutting machine and first floor water cooler cables were damaged. Damaged insulation compromises current flow, potentially causing overheating. Persistent overheating degrades cable quality and raises fire risk.

For any future electrical work, the termination of stranded cables must be done with cable lugs and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202603-063835

Timeline	Remediation Plan
Medium Term CAP-0074727	i. Install cable lugs on all improperly terminated cables within the distribution boards to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0074728	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.

Medium Term CAP-0074729	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with BS7671 18th Edition + A2 (2022).
Longer Term CAP-0074730	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Longer Term CAP-0074731	iii. Repair or replace the damaged cable insulation to restore proper insulation, in compliance with BS 7671, Regulation 526.1, which requires all connections to be electrically sound and insulated to prevent accidental contact.



Figure 2.50 - PVC tape joint was found at main building rooftop lift room



Figure 2.51 - PVC tape joint was found at kitchen area



Figure 2.52 - Main building basement cutting machine cables were found damaged



Figure 2.53 - Main building basement cutting machine cables were found damaged

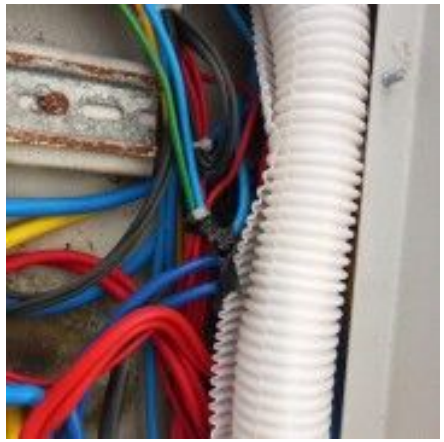


Figure 2.54 - PVC tape joint was found at main building ground floor distribution board

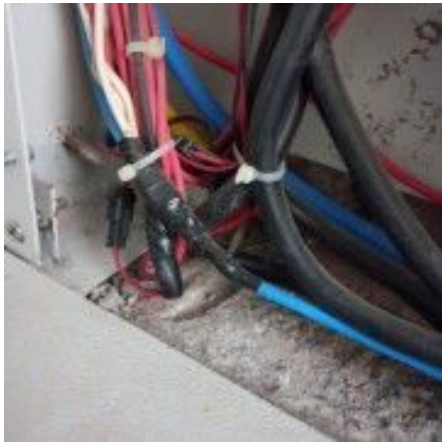


Figure 2.55 - PVC tape joint was found at main building first floor distribution board



Figure 2.56 - Cables were terminated without cable lugs at main building first floor production area distribution board



Figure 2.58 - Cables were terminated without cable lugs at office building first floor production area distribution board



Figure 2.57 - Main building first floor water cooler cable was found damaged due to switch board

11.3 Colour coding

This section relates to entire facility.

During the inspection of multiple distribution boards, deviations from standard cable colour coding practices were identified.
Observed at,
a) Main building ground floor LT room panels.
b) Main building basement area distribution board.

Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding. This will ensure the accurate identification of conductors, thereby enhancing overall safety and operational efficiency.F-202603-063837

Timeline	Remediation Plan
Medium Term CAP-0074723	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labeling or re-insulating the cables. The color of the cables should be maintained as per the phase it is connected to.
Longer Term CAP-0074724	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices.



Figure 2.59 - Colour coding violation was found at main building ground floor LT room panel



Figure 2.61 - Colour coding violation was found at main building basement distribution board



Figure 2.60 - Colour coding violation was found at main building ground floor LT room solar panel

11.4 Fixtures

This section relates to the entire facility.
During the inspection, the electrical fixtures throughout the facility were visually surveyed.

It was observed that the wooden backing was installed with a switchboard at many locations. Wood is flammable, and if the electric circuit inside the wooden switchboard generates excessive heat due to overloads or faults, it can ignite the wooden box. This can lead to a fire hazard.
Observed at,
a) Main building first floor mechanical workshop.
b) Ground floor main gate fire control room.F-202603-063838

Timeline	Remediation Plan
Medium Term CAP-0074721	Replace the wooden backing from switchboard with a non-combustible enclosure designed for electrical installations.
Longer Term CAP-0074722	Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.



Figure 2.62 - Wooden switchboard was found at main building first floor mechanical workshop room



Figure 2.63 - Wooden switch board was found at ground floor fire control room

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.